Design md

Team members: Cesar Ernesto Perez Gomez
Iker Jesus Ortiz Rojero
Rodrigo Zatarain Aguirre

Three interactions decisions:

Interaction decision 1: A large confirmation button in the calendar screens.
User problem: Users may have difficulty selecting a small button, especially when interacting with a touchscreen kiosk. Small targets require more precise movements and can lead to missed taps or accidental selections.

HCI principle: Fitts' Law. The time needed to reach a target depends on its distance and size. Increasing the target size can make selection easier without requiring such precise movements.

Design decision and justification: We placed a wide "CONFIRM RESERVATION" button near the bottom of the calendar screen. Its size provides a large clickable area, while its position separates the main action from the date and time selections. This design aims to make the button easier to locate and select with either a mouse or a finger.

Image:

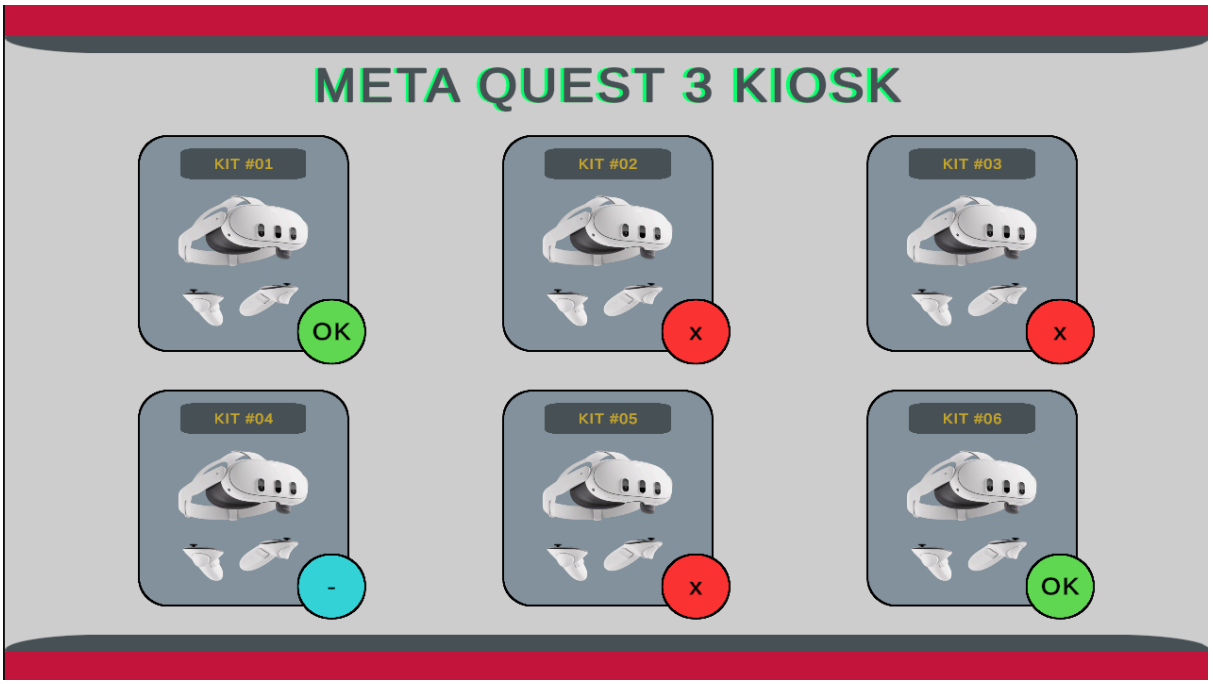


Interaction decision 2: Visual indicators for kit availability
User problem: Users need to identify which Meta Quest kits are available before choosing one. Opening each kit individually or reading detailed descriptions would require additional effort and slow down the selection process.

HCI principle: Norman's Signifiers. Visible cues communicate information that helps users understand an interface and identify possible actions.

Design decision and justification: The kit selection screen displays status indicators beside each kit. A green circle with a "OK" text represents availability, while a red cross indicates that a kit is unavailable. The design combines symbols with colors so that users have more than one visual cue to interpret the status.

Image:



Interaction decision 3: Enabling confirmation only after a valid time selection
User problem: Users may try to continue without selecting a time or attempt to choose an unavailable slot. Allowing these actions would leave the next step without the information required to continue correctly.
HCI principle: Norman's Constraints and error prevention. Restricting actions that are not currently valid guides the interaction and prevents users from entering an invalid state.

Design decision and justification: The calendar initially displays the confirmation button with reduced opacity and disables interaction with it. In the calendars of Scenarios A and B.2, selecting the available time slot enables the button and restores its full opacity. Unavailable time slots remain disabled. In the calendar of Scenario B.1, both time slots are unavailable, so confirmation remains disabled. Users can select day 28 to open the calendar of Scenario B.2 and see the alternative time. This decision makes a valid time selection a requirement before continuing. The change in the button's appearance also provides feedback that the selection has been accepted.

Image:



Comparison between Unity and Figma

The main difference lies in the transition from a purely graphic (static) design to a real-time functional architecture (GameObjects and UI Engine).
In Figma, the elements are flat visual layers (vectors, boxes, and drawn text). Their purpose is to define the aesthetics, proportions, color palette, and visual layout. In Unity: Each element becomes a GameObject within a Canvas, featuring specific engine components.

In Figma, The information is "frozen" within the design (for example, the text "KIT #06", "22/September/2026", or "KIT #01" consists of static, illustrative text).
In Unity,the display is decoupled so that these fields do not rely on a static image but can instead be updated in real time via variables or scripts (dynamically showing which kit was assigned to the user, the delivery date, or the reservation status).

And we also added a “Home button” in the last screen of both scenarios of our application, when the user borrows a quest, and clicks that button, he gets to the first screen of our application but the quest that he just borrowed now appears that is not available, because of this our application saves the loan and updates the inventory correctly.

Changes made after implementation requirements, usability findings or testing evidence.

We made two changes to our project after testing evidence made to fulfill one requirement of the **Testing.md** document. The first change we made where to the three calendar scenes of our project, this change was that we made bigger in scale the labels of the current time and the available times of each day, we made them so the user can have more visible clarity on the day that he is making the reservation and which times are available in a certain day.
The second change was to the Confirm Reservation button on our three calendar scenes. We changed the color palette so it could be more visible to the user while navigating through the calendar scene.